

Experiment No : 06

Experiment Name : Study of single stage FET amplifier circuit

Objectives:

- To implement a FET amplifier circuit
- To observe the input/output waveforms of a FET amplifier circuit

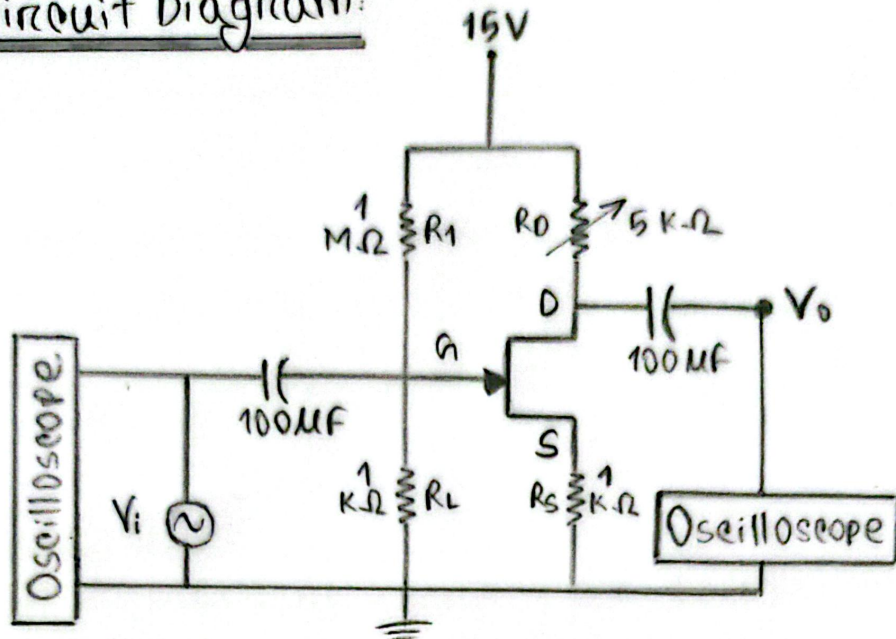
Theory:

A Field Effect Transistor is a voltage controlled semiconductor device that operates with high input impedance and low output impedance, making it suitable for amplification applications. A common source JFET amplifier has high input impedance and a moderate voltage gain. Also, the input and output voltages are 180° out of phase.

Required Apparatus:

- JFET (2N5458, 1 piece)
- Resistors ($1\text{ M}\Omega$: 1 piece,
 $1\text{ K}\Omega$: 2 piece)
- Potentiometer ($100\text{ K}\Omega$: 1 piece)
- Capacitors ($10\text{ }\mu\text{F}$: 1 piece
 $100\text{ }\mu\text{F}$: 2 piece)
- DC power supply (0-15V)
- Signal generator
- Oscilloscope
- Project Board
- Multimeter
- Connecting wires

Circuit Diagram:



Procedure:

- (i) The circuit was connected according to the figure
- (ii) Mid-point biasing was performed
- (iii) The coupling capacitors and bypass capacitors were connected
- (iv) A small signal to the input side was supplied using a signal generator and the corresponding output waveform was recorded
- (v) The experimental voltage gain of the amplifier circuit was compared with the theoretical gain

Calculations:

Theoretical, $A_v = -g_m (R_D \parallel R_L) = -g_m R_D = 3.5$

Experimental, $A_v' = \frac{V_o (p-p)}{V_i (p-p)}$
 $= \frac{3.5}{1} = 3.5$

$$g_m = \frac{A_v'}{R_D} = \frac{3.5}{5 \times 10^3} = 0.7 \text{ mS}$$

Discussion and Conclusion:

In this experiment, the single-stage common source JFET amplifier was successfully implemented and the output waveform showed 180° phase shift as expected. Thus, the amplification characteristics of the FET amplifier were verified.

Precautions:

- All connections were given correctly
- Transistor terminals were identified properly
- Calibration was performed accurately